

Section 8.1 Worksheet: Complex Numbers

Instructions

Show all algebraic steps. Write each final complex number in standard form $a + bi$.

1. For $z = 5 - 2i$ and $w = -1 + 4i$, compute:

$$z + w, \quad z - w, \quad 3z - 2w, \quad zw.$$

2. Plot z , w , and $z + w$ in the complex plane. Explain why the point for $z + w$ is obtained by vector addition.

3. Solve

$$x^2 - 8x + 25 = 0$$

and verify both roots by substitution.

4. Let

$$A = \begin{bmatrix} 2+i & 1 \\ -i & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2i \\ 2 & 1-i \end{bmatrix}.$$

Compute $A + B$, AB , and $\det(A)$.

5. Compute BA for the matrices in Problem 4. Compare AB and BA and explain why the products differ.
6. A fictional R&B audio component is represented by $s = 3 + 4i$. What are its real and imaginary components? What point and vector represent it in the complex plane?

Section 8.2 Worksheet: Conjugates, Modulus, and Division

Instructions

Use conjugates explicitly. For matrix inverses, verify by multiplication.

1. For $z = -3 + 4i$, find $\bar{z}$, $z\bar{z}$, and $|z|$.

2. Verify directly that

$$\overline{z + w} = \bar{z} + \bar{w}$$

for $z = 2 - 5i$ and $w = -1 + 3i$.

3. Simplify

$$\frac{7 - 3i}{1 + 2i}$$

to standard form.

4. Prove that

$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$$

for $w \neq 0$.

5. Find and verify the inverse of

$$A = \begin{bmatrix} 2 & i \\ -i & 1 \end{bmatrix}.$$

6. Show that for $z \neq 0$,

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Then use the formula to find $(3 - 4i)^{-1}$.

Section 8.3 Worksheet: Polar Form and De Moivre's Theorem

Instructions

Use exact angles whenever possible. Identify the quadrant before choosing an argument.

1. Convert $z = -1 - \sqrt{3}i$ to polar form using the principal argument.

2. Convert

$$4 \operatorname{cis} \left(\frac{5\pi}{6} \right)$$

to standard form.

3. Let

$$z_1 = 3 \operatorname{cis} \left(\frac{\pi}{3} \right), \quad z_2 = 2 \operatorname{cis} \left(-\frac{\pi}{6} \right).$$

Find $z_1 z_2$ and z_1 / z_2 in polar and standard forms.

4. Use De Moivre's theorem to compute $(\sqrt{3} + i)^6$.

5. Find all cube roots of -8 and verify one root directly.

6. Explain geometrically why the n th roots of a nonzero complex number are equally spaced on a circle.

Section 8.4 Worksheet: Complex Vector Spaces and Inner Products

Instructions

Use the convention $\langle \mathbf{u}, \mathbf{v} \rangle = \sum u_j \overline{v_j}$, linear in the first argument.

1. For $\mathbf{u} = (1 + i, 2 - i)$ and $\mathbf{v} = (2, -i)$, compute $\mathbf{u} + \mathbf{v}$ and $(1 - 2i)\mathbf{u}$.
2. Determine whether $\{(1, i), (i, 1)\}$ is a basis for $\mathbb{C}^2$.
3. Using the vectors in Problem 1, compute

$$\langle \mathbf{u}, \mathbf{v} \rangle, \quad \|\mathbf{u}\|, \quad d(\mathbf{u}, \mathbf{v}).$$

4. Find the projection of $\mathbf{u} = (1 + i, 2)$ onto $\mathbf{v} = (1, -i)$ and verify that the residual is orthogonal to $\mathbf{v}$.
5. Apply one step of complex Gram–Schmidt to

$$\mathbf{v}_1 = (1, i), \quad \mathbf{v}_2 = (1, 1).$$

6. Explain with a specific nonzero vector why the formula $\sum u_j v_j$ is not a valid positive-definite inner product on $\mathbb{C}^n$.

Section 8.5 Worksheet: Unitary and Hermitian Matrices

Instructions

Show conjugation and transposition separately when computing A^* . For diagonalization, normalize every eigenvector.

1. Find A^* for

$$A = \begin{bmatrix} 1+i & 2 \\ 3i & -1+i \end{bmatrix}.$$

2. Prove directly, entry by entry, that $(cA)^* = \bar{c}A^*$.

3. Determine whether

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

is unitary.

4. Verify that

$$H = \begin{bmatrix} 3 & 1+i \\ 1-i & 3 \end{bmatrix}$$

is Hermitian and find its eigenvalues.

5. Unitarily diagonalize

$$H = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}.$$

6. Prove that if U is unitary, then $\|U\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{C}^n$.